SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*

Assessment Tool Weightage: 40%

Annexure A

SHORT ANSWER TEST

Code of Conduct:

I Devi Shree Babu (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Short Answer Test

1. A company needs to transfer a **150 MB** design file from a client computer to a remote server over a network with a bandwidth of **50 Mbps**. The Transport Layer divides the file into **1500-byte segments**, while the Data Link Layer encapsulates each segment into frames by adding **40 bytes of header and trailer**. Calculate the total number of segments and frames transmitted, the total transmission overhead introduced by the Data Link Layer, and estimate

the total transmission time (ignoring propagation delay). Explain how the Transport and Data Link layers ensure reliable communication.

Answer:

Given

- File size (F) = 150 MB = 150,000,000 bytes
- Segment size (S) = 1500 bytes
- Data Link Layer overhead per frame (H) = 40 bytes
- Bandwidth (B) = 50 Mbps = 50×10^6 bits/s

1. Number of Segments

Formula:

Number of Segments = File Size / Segment Size

= 150,000,000 / 1500

= 100,000 segments

2. Number of Frames

Formula:

Number of Frames = Number of Segments

= 100,000 frames

3. Total Transmission Overhead

Formula:

Overhead = Number of Frames × Overhead per Frame

= 100,000 × 40

= 4,000,000 bytes

= 4 MB

4. Total Transmission Time

Formula:

Transmission Time = ((File Size + Overhead) × 8) / Bandwidth

= ((150,000,000 + 4,000,000) × 8) / 50,000,000

= 1,232,000,000 / 50,000,000

= 24.64 seconds

5. Reliable Communication

- **Transport Layer:** Divides the file into segments and ensures reliable delivery using sequencing, acknowledgments (ACKs), and retransmissions.
- **Data Link Layer:** Encapsulates segments into frames, detects errors using CRC, and retransmits corrupted frames to ensure reliable communication.

2. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

Answer:

Given

- Sliding window size (W) = 6 frames
- Data per frame = 1024 bytes
- Total frames transmitted (F) = 48 frames

1. Number of Transmission Windows

Formula:

Number of Windows = Total Frames / Window Size

= 48 / 6

= 8 windows

2. Total Number of Retransmissions

Since one frame is lost in every window,

Formula:

Total Retransmissions = Number of Windows × Lost Frames per Window

= 8 × 1

= 8 frames

How Flow Control Improves Communication Efficiency ?

- Flow control ensures that the sender does not transmit data faster than the receiver can process it.
- It prevents buffer overflow at the receiver.
- It reduces packet loss and unnecessary retransmissions.
- It improves network efficiency and reliable data transfer by matching the transmission rate with the receiver's capacity.

3. A **12,000-byte** IP packet is transmitted through a router where the Maximum Transmission Unit (MTU) is **1500 bytes**. Assuming each fragment requires a **20- byte IP header**, calculate the total number of fragments created and the total header overhead introduced. Explain how the Network Layer ensures successful delivery of fragmented packets to the destination.

Answer:

Given

- IP packet size (P) = 12,000 bytes
- MTU = 1,500 bytes

- IP header per fragment (H) = 20 bytes

1. Payload per Fragment

Formula:

Payload per Fragment = MTU − IP Header

= 1500 − 20

= 1480 bytes

2. Number of Fragments

Formula:

Number of Fragments = Ceiling(Packet Size / Payload per Fragment)

= Ceiling(12,000 / 1480)

= Ceiling(8.108)

= 9 fragments

3. Total Header Overhead

Formula:

Total Header Overhead = Number of Fragments × IP Header Size

= 9 × 20

= 180 bytes

Network Layer

The Network Layer fragments large packets, forwards each fragment independently, and reassembles them at the destination using the fragment identification and offset fields.

4. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries

1024 bytes of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

Answer:

Given

- Sliding window size (W) = 6 frames
- Data per frame = 1024 bytes
- Total frames transmitted (F) = 48 frames

1. Number of Transmission Windows

Formula:

Number of Windows = Total Frames / Window Size

= 48 / 6

= 8 windows

2. Total Number of Retransmissions

Since one frame is lost in every window,

Formula:

Total Retransmissions = Number of Windows × Lost Frames per Window

= 8 × 1

= 8 frames

Flow Control :

Flow control ensures that the sender transmits data at a rate the receiver can handle, preventing buffer overflow, reducing retransmissions, and improving communication efficiency.

5. A multimedia application transfers a **600 MB** video file. The Session Layer inserts a synchronization checkpoint after every **60 MB** of transmitted data. During transmission, a failure occurs after **390 MB** has been transferred. Determine the number of checkpoints created before failure and calculate the amount of data that must be retransmitted after recovery. Explain the importance of synchronization in the Session Layer.

Answer:

Given

- Video file size = 600 MB
- Checkpoint interval = 60 MB
- Failure occurs after = 390 MB

1. Number of Checkpoints Before Failure

Formula:

Number of Checkpoints = Data Transferred / Checkpoint Interval

= 390 / 60

= 6 checkpoints

2. Data to be Retransmitted

Last checkpoint = 360 MB

Formula:

Data to Retransmit = Data Transferred − Last Checkpoint

= 390 − 360

= 30 MB

Importance of Synchronization

The Session Layer uses synchronization checkpoints to allow communication to resume from the last saved checkpoint after a failure, reducing retransmission time and improving efficiency.

6. Before transmission, the Presentation Layer compresses a **900 MB** file by **40%**. The compressed data is encrypted, adding **5%** overhead to the compressed size. Calculate the compressed file size, the encrypted file size, and the total percentage reduction compared to the original file. Explain how the Presentation Layer improves transmission efficiency and security.

Answer:

Given

- Original file size = 900 MB
- Compression = 40%
- Encryption overhead = 5% of compressed size

1. Compressed File Size

Formula:

Compressed Size = Original Size × (1 − Compression %)
= 900 × (1 − 0.40)
= 900 × 0.60
= 540 MB

2. Encrypted File Size

Formula:

Encrypted Size = Compressed Size × (1 + Encryption Overhead %)
= 540 × 1.05
= 567 MB

3. Total Percentage Reduction

Formula:

Percentage Reduction = ((Original Size − Encrypted Size) / Original Size) × 100
= ((900 − 567) / 900) × 100
= (333 / 900) × 100
= 37%

Presentation Layer

The Presentation Layer improves transmission efficiency by compressing data to reduce its size and enhances security by encrypting the data before transmission.

7.An FTP application transfers **3 GB** of data over a network with an effective throughput of **120 Mbps**. The Application Layer divides the data into requests of **3 MB** each. Calculate the total number of requests generated and estimate the total transmission time in seconds. Discuss how the Application Layer provides reliable services to users.

Answer:

Given

- Data size = 3 GB = 3000 MB
- Request size = 3 MB
- Throughput = 120 Mbps

1. Total Number of Requests

Formula:

Number of Requests = Total Data / Request Size
= 3000 / 3
= 1000 requests

2. Total Transmission Time

Convert data to bits:

3000 MB × 8 = 24,000 Mb

Formula:

Transmission Time = Total Data (Mb) / Throughput (Mbps)
= 24,000 / 120
= 200 seconds

Application Layer

The Application Layer provides services such as FTP for file transfer, manages user requests, ensures reliable data exchange, and allows applications to communicate efficiently over the network.

8.A packet experiences the following processing delays at different OSI layers: Application Layer = **4 ms**, Transport Layer = **3 ms**, Network Layer = **5 ms**, Data Link Layer = **2 ms**, and Physical Layer = **1 ms**. The propagation delay is **18 ms**, and the transmission delay is **12 ms**. Calculate the total end-to-end delay experienced by the packet and explain how each layer contributes to the communication process.

Answer:

Given

- Application Layer delay = 4 ms
- Transport Layer delay = 3 ms
- Network Layer delay = 5 ms
- Data Link Layer delay = 2 ms
- Physical Layer delay = 1 ms
- Propagation delay = 18 ms
- Transmission delay = 12 ms

1. Total Processing Delay

Formula:

Processing Delay = Application + Transport + Network + Data Link + Physical
= 4 + 3 + 5 + 2 + 1
= 15 ms

2. Total End-to-End Delay

Formula:

Total Delay = Processing Delay + Propagation Delay + Transmission Delay
= 15 + 18 + 12
= 45 ms

Contribution of Each Layer

- Application Layer: Provides network services to applications.
- Transport Layer: Ensures reliable data delivery and flow control.
- Network Layer: Determines the routing path for packets.
- Data Link Layer: Performs framing and error detection.
- Physical Layer: Transmits data as electrical, optical, or radio signals over the communication medium.

9.A network transfers **80 MB** of useful data using frames of **1600 bytes**, where each frame contains **80 bytes** of protocol overhead added by different OSI layers.
Calculate the total number of frames required, the total overhead transmitted, and the transmission efficiency as a percentage. Explain how protocol overhead affects network performance.
Answer:

Given

- Useful data = 80 MB = 80,000,000 bytes
- Frame size = 1600 bytes
- Protocol overhead per frame = 80 bytes

1. Payload per Frame

Formula:

Payload per Frame = Frame Size − Overhead
= 1600 − 80
= 1520 bytes

2. Total Number of Frames

Formula:

Number of Frames = Ceiling(Useful Data / Payload per Frame)
= Ceiling(80,000,000 / 1520)
= 52,632 frames

3. Total Overhead Transmitted

Formula:

Total Overhead = Number of Frames × Overhead per Frame
= 52,632 × 80
= 4,210,560 bytes
≈ 4.21 MB

4. Transmission Efficiency

Formula:

Efficiency = (Useful Data / (Useful Data + Overhead)) × 100
= (80,000,000 / (80,000,000 + 4,210,560)) × 100
= (80,000,000 / 84,210,560) × 100
≈ 94.99%
≈ 95%

Protocol Overhead

Protocol overhead adds extra header and trailer information to each frame for addressing, error detection, and control. While it ensures reliable communication, it increases the amount of transmitted data, reducing overall network efficiency.

10.A hospital transfers a patient's medical record of **240 MB** through an IoT enabled healthcare network. The Presentation Layer compresses the data by **30%**, the Transport Layer divides the compressed data into **1400-byte segments**, and the Data Link Layer adds **60 bytes** of overhead to each frame. Calculate the compressed file size, the total number of segments transmitted, the total protocol overhead introduced, and the final amount of data transmitted across the Physical Layer. Explain how multiple OSI layers work together to ensure secure and reliable communication.

Answer:

Given

- Original file size = 240 MB = 240,000,000 bytes
- Compression = 30%
- Segment size = 1400 bytes
- Data Link Layer overhead = 60 bytes/frame

1. Compressed File Size

Formula:

Compressed Size = Original Size × (1 − Compression %)
= 240 × 0.70
= 168 MB
= 168,000,000 bytes

2. Total Number of Segments

Formula:

Number of Segments = Compressed Size / Segment Size
= 168,000,000 / 1400
= 120,000 segments

3. Total Protocol Overhead

Formula:

Protocol Overhead = Number of Segments × Overhead per Frame
= 120,000 × 60
= 7,200,000 bytes
= 7.2 MB

4. Final Data Transmitted

Formula:

Final Data = Compressed Size + Protocol Overhead
= 168,000,000 + 7,200,000
= 175,200,000 bytes
= 175.2 MB

Multiple OSI Layers

- Presentation Layer: Compresses data to reduce size and improve efficiency.
- Transport Layer: Divides data into segments and ensures reliable delivery.
- Data Link Layer: Adds headers/trailers for framing and error detection.
- Physical Layer: Transmits the final data over the communication medium, ensuring successful communication.

RUBRICS FOR CALCULATION

Criteria	Weightag e	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understand in g of Concepts	25%	Demonstrat es a thorough, accurate understan din g of concepts	Good understandi ng with minor gaps	Basic understandi ng with some misconcepti on s	Poor understandin g; major misconcepti on s
Explanation	25%	Detailed, well explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organizatio n, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, profession	Understand abl e with	Limited clarity,	Poorly written,

		al presentation	minor issues	readability issues	difficult to understand